

# Skills Summary

## Algebraic Expressions — Distributing and Combining Safely

*Use this reference while completing your worksheet or when studying.*

### Skill 1 — Distribute to every term, sign included

**Rule:**  $k(p + q) = kp + kq$  — the multiplier outside reaches *every* term inside, and it keeps its own sign while doing it.

**Signs:** negative times negative is positive, so  $-2(x - 3) = -2x + 6$ . Write the sign as part of the number you are distributing, and the second term takes care of itself.

**The substitution test:** put any number in both versions. Equivalent expressions always agree; one disagreement proves the step was illegal.

**Example:** Write  $6(x - 4)$  without parentheses.

**Step 1.** Two terms sit inside the parentheses, so distributing must produce two products — the subtraction sign belongs to the second one.

**Step 2.**  $6 \cdot x = 6x$  and  $6 \cdot (-4) = -24$ , and at  $x = 5$  both versions give 6, which is the check.  
 **$6x - 24$**

**Try it:** Write  $-2(x + 7)$  without parentheses.

check:  $-2x - 14$

### Skill 2 — Combine only like terms

**Like terms** have exactly the same letter part:  $7a$  and  $-2a$  are like;  $5x$  and  $3y$  are not, and  $5a$  and 13 are not.

**Rule:** add or subtract the coefficients and keep the letter part unchanged —  $7a - 2a = 5a$ , never  $5a^2$  and never 5.

**Unlike terms simply stay put.**  $5x + 3y$  is already simplified; an expression is allowed to have more than one term, and forcing it into one is the most common error on this page.

**Example:** Simplify  $4b + 7 - b + 2$ .

**Step 1.** Sort the terms into groups with the same letter part first, carrying each sign with its own term, so nothing unlike ever meets.

**Step 2.** The  $b$  terms give  $4b - b = 3b$ ; the constants give  $7 + 2 = 9$ ; and  $3b$  and 9 are unlike, so the work stops there.  **$3b + 9$**

**Try it:** Simplify  $9m - 3 - 4m + 8$ .

check:  $5m + 5$

### Skill 3 — Factor out, then distribute back

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**Factoring is distributing in reverse:** find the greatest common factor of the terms, write it outside, and write what is left inside.

$kp + kq = k(p + q)$ . Divide EVERY term by the common factor — leaving one term undivided is the same mistake as distributing to only one term.

**Always check:** distribute your answer back. If it does not rebuild the original expression exactly, the factoring is wrong and you will see it at once.

**Example:** Factor  $8x + 20$ .

**Step 1.** Both terms share a numerical factor, so the greatest common factor comes out front and each term is divided by it.

**Step 2.** The common factor is 4; dividing gives  $8x \div 4 = 2x$  and  $20 \div 4 = 5$ , and distributing back returns  $8x + 20$  ✓.  $4(2x + 5)$

**Try it:** Factor  $9x + 12$ , then distribute back to check.

check:  $(4 + x)3$

**(!) Watch out:** the two errors are opposites. Distributing to only the first term does too little; merging unlike terms does too much. One substitution catches either one in about ten seconds.